

Isaac Mubiru

isaacmubiru99@gmail.com | linkedin.com/in/isaac-mubiru | github.com/zac-09 | Kampala, Uganda

PROFESSIONAL SUMMARY

Backend engineer with 6+ years building **microservices, event-driven data flows and third-party integrations** on production systems with sizable user bases. Led the end-to-end migration of **20+ production applications** to Firebase/GCP with zero downtime, reporting directly to the CTO; built a live two-way MongoDB↔Firestore pipeline on **Google Pub/Sub** that kept data consistent under parallel production traffic; integrated **Morningstar market-data APIs** across 2M+ records for a live investing app; and architected an event-driven backend on **Apache Kafka, Docker and Kubernetes**. Production depth in Node.js/TypeScript, with **Python** as a second language (migration tooling, SDK work) and intermediate **Go**. Looking to apply the same ownership-from-infancy-to-production approach to a wealth platform used by thousands.

CORE COMPETENCIES

Microservices & Event Sourcing

REST & gRPC API Design

3rd-Party API Integration

Data Pipelines & Consistency

Python & Node.js

Docker & Kubernetes

GCP & AWS

Zero-Downtime Migrations

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer · US Remote

- Owned the end-to-end migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, from initial design through production cutover, reporting directly to the CTO.
- Built a real-time two-way sync between MongoDB and Firestore using **Node.js** and Google **Pub/Sub** for message processing — an event-driven data flow that kept both stores consistent while serving live production traffic.
- Migrated file storage from Amazon S3 to Google Cloud Storage with a **Python** migration pipeline.
- Integrated third-party platforms into the new backend: moved a Webflow website off Parse onto Firebase; built a 3D visualisation feature (video overlay + ffmpeg) that increased buyer conversion.
- Shipped Express Shipping end to end (+\$40 revenue per order) and cut infrastructure cost by **\$5,000/month** by consolidating all services off AWS.

MTailor

Mar 2022 – Jul 2022

Backend Contractor · US Remote

- Built the migration pipeline that read from MongoDB, applied complex processing logic, and wrote to Firestore for a full backend migration.
- Wrote documentation for fellow engineers on the new Firebase SDKs (**Node.js and Python**) and trained the Ops team on the Firebase dashboard and backup procedures.

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer · Singapore Remote

- Integrated **Morningstar APIs** into a live investing PWA: built scheduled jobs querying **over 2 million Firestore records** to keep customer-facing stock prices current.
- Extended backend APIs in **TypeScript** using Firebase Cloud Functions and **Express.js** on Heroku; led optimisation of the real-time price delivery path.

Mind2matter

Jan 2022 – Jul 2022

Software Engineer · US Remote

- Built backends for DeFi applications using **Node.js** and Web3 — value-moving transaction flows and APIs under tight client timelines, reporting directly to the CTO.

CodeBits

Jan 2020 – Jul 2021

Software Engineer / Team Lead · Uganda Remote

- Architected an event-driven **microservices** backend for FIDA Uganda's case-management platform using **Apache Kafka, Docker and Kubernetes**, with NATS Streaming for service messaging.
- Built a USSD legal-aid service communicating with a **Node.js** backend via **gRPC**; delivered a paralegal database for LASPNET Uganda.
- Led a team of 4 developers; shipped a cross-platform React Native (Expo) app with Firebase push notifications.

PROJECTS

Real-time Pub/Sub Sync Pipeline

Two-way MongoDB↔Firestore event pipeline with idempotent, redelivery-safe message handling, running under parallel production traffic across 20+ applications during a zero-downtime migration.

Node.js · Google Pub/Sub · MongoDB · Firestore

Morningstar Market-Data Integration

Scheduled jobs keeping 2M+ Firestore records current from external Morningstar APIs for a live customer-facing investing PWA.

Node.js · TypeScript · Firestore · Cloud Functions · External API integration

Event-Driven Microservices — FIDA Uganda

Kafka-based event-driven backend for a legal case-management platform, containerised with Docker and orchestrated on Kubernetes; gRPC service contract for a USSD gateway.

Apache Kafka · Docker · Kubernetes · gRPC · NATS Streaming

Cross-Cloud Storage Migration

Python pipeline migrating object storage from Amazon S3 to Google Cloud Storage as part of a consolidation that saved \$5,000/month.

Python · AWS S3 · Google Cloud Storage

EDUCATION

B.Sc. Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

LINKS

Open-source & project code — github.com/zac-09

SKILLS

Languages: JavaScript/Node.js, TypeScript, Python, SQL, Go (intermediate), Rust (intermediate)

Backend & Architecture: Microservices, Event-driven systems, REST APIs, gRPC, Express.js, Firebase Cloud Functions

Messaging & Data: Apache Kafka, Google Pub/Sub, NATS Streaming, MongoDB, Firestore

Cloud / Infra: GCP (Firestore, Pub/Sub, Cloud Functions, GCS), AWS (S3, EC2, EBS — intermediate), Docker, Kubernetes, Linux, Git

Spoken: English, Kiswahili